

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1156

Dowell 138 kV
April 02, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1156. The Interconnection Customer proposes a 111.6 MW battery energy storage facility interconnecting at the Dowell 138 kV substation in the Kentucky area. The request was received on February 22, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 434 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$801,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SIGGEN - HYAMPOM 138 kV line	HUBBARDSTOWN - TRISTATE 345 kV	115.1	395	6.4
LOVELAND - BERRY ST 138 kV L	BERRY STREET - NORTHFIELD 345 kV	107.2	328	6.8
OJO SUBSTATION - LOVELAND 138 kV	BERRY STREET - NORTHFIELD 345 kV	107.0	332	6.8
DOWELL - WEST PAINTSVILLE 138 kV Line	WISSOTA - NSP 138 kV	105.4	427	65.1
HYAMPOM - HUBBARDSTOWN SUBSTATION 138 kV	TRISTATE SUBSTATION - HUBBARDSTOWN 345 kV	104.2	484	7.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SIGGEN - HYAMPOM 138 kV line	Rebuild line with higher-capacity conductor	\$3,069,000	4% (\$122,000)
LOVELAND - BERRY STREET 138 kV	Reconductor line	\$5,152,000	5% (\$255,000)
LOVELAND - OJO 138 kV	Reconductor line	\$2,723,000	5% (\$135,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DOWELL - WEST PAINTSVILLE 138 kV Line	Reconductor line	\$1,070,000	25% (\$265,000)
HYAMPOM - HUBBARDSTOWN SUBSTATION 138 kV	Reconductor line	\$577,000	4% (\$24,000)
	Total allocated to project		\$801,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.